

BUSINESS INTELLIGENCE DASHBOARD DEVELOPMENT USING POWER BI

Internship Project Report:

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ABSTRACT :

Business Intelligence (BI) enables organizations to transform raw data into meaningful information for better decision-making. This project focuses on designing and developing an interactive Business Intelligence dashboard using Microsoft Power BI. A sales dataset was imported, cleaned, transformed, and modeled to create an efficient reporting system.

The project involved data preprocessing using Power Query Editor, creation of DAX measures for business calculations, development of a star schema, and construction of interactive dashboards. Various visualizations such as KPI cards, bar charts, line charts, pie charts, donut charts, treemaps, and slicers were used to present business insights clearly.

The completed dashboard allows users to analyze sales performance, customer distribution, product categories, and sales trends over time, helping businesses make informed decisions.

INTRODUCTION:

Business Intelligence (BI) is the process of collecting, analyzing, and visualizing data to support business decision-making. Organizations generate large amounts of transactional data every day, but raw data alone does not provide meaningful insights. Business Intelligence tools help convert raw information into interactive reports and dashboards.

Microsoft Power BI is one of the most widely used Business Intelligence tools because it enables users to connect multiple data sources, transform data, perform calculations using DAX, and create interactive dashboards. It provides businesses with a simple yet powerful platform for data analysis.

In this project, a sales dataset was analyzed to identify business trends, customer behavior, product performance, and geographical sales distribution. The dashboard developed in Power BI provides an interactive interface for exploring business performance efficiently.

OBJECTIVES:

The objectives of this project are:

- To understand Business Intelligence concepts.
- To import and preprocess sales data using Power Query.
- To clean and transform data for analysis.
- To develop a star schema for efficient data modeling.
- To create DAX measures for business calculations.
- To design interactive dashboards using Power BI.
- To analyze sales performance through different visualizations.
- To support business decision-making using data insights.

TOOLS AND TECHNOLOGIES:

Tool	Purpose
Microsoft Power BI Desktop	Dashboard Development
Power Query Editor	Data Cleaning
DAX	Business Calculations
CSV Dataset	Data Source

Microsoft Excel

Dataset
Preparation

GitHub

Project Repositor

DATASET DESCRIPTION:

The project uses the **Sales Data Sample** dataset containing sales transactions from different customers across multiple countries.

Important Fields

- Order Number
- Order Date
- Customer Name
- Product Line
- Country
- Sales
- Deal Size
- Order Status
- Quantity Ordered
- Price Each

The dataset contains sales information that can be analyzed to understand customer purchasing behavior and business performance.

DATA PREPROCESSING:

Data preprocessing is one of the most important steps in Business Intelligence because raw data often contains inconsistencies, incorrect data types, and unnecessary information. Before creating dashboards, the

dataset was cleaned and transformed using **Power Query Editor** in Microsoft Power BI.

The following preprocessing steps were performed:

- Imported the **Sales Data Sample** CSV dataset into Power BI.
- Verified the structure and quality of the dataset.
- Converted the **ORDERDATE** column to the Date/Time data type.
- Checked all numeric columns such as Sales, Quantity Ordered, and Price Each.
- Ensured text columns like Customer Name, Product Line, Country, and Status were correctly formatted.
- Loaded the cleaned dataset into the Power BI data model.

Table.TransformColumnTypes(*Promoted Headers*,({("ORDERNUMBER", Int64.Type), ("QUANTITYORDERED", Int64.Type), ("PRICEEACH", type number), ("ORDERLINENUMBER", Int64.Type), ("SALES", type number), ("ORDERDATE", type date-time), ("STATUS", type text)}))

	1	2	3	4	5	6	7
	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDERDATE	STATUS
1	10107	30	95.7	2	2871	2/24/2003 0:00	Shipped
2	10121	34	81.35	5	2765.9	5/7/2003 0:00	Shipped
3	10134	41	94.74	2	3884.34	7/1/2003 0:00	Shipped
4	10145	45	83.26	6	3746.7	8/25/2003 0:00	Shipped
5	10159	49	100	14	5205.27	10/10/2003 0:00	Shipped
6	10168	36	96.66	1	3479.76	10/28/2003 0:00	Shipped
7	10180	29	86.13	9	2497.77	11/11/2003 0:00	Shipped
8	10188	48	100	1	5512.32	11/18/2003 0:00	Shipped
9	10201	22	98.57	2	2168.54	12/1/2003 0:00	Shipped
10	10211	41	100	14	4708.44	1/15/2004 0:00	Shipped
11	10223	37	100	1	3965.66	2/20/2004 0:00	Shipped
12	10237	23	100	7	2333.12	4/5/2004 0:00	Shipped
13	10251	28	100	2	3188.64	5/18/2004 0:00	Shipped
14	10263	34	100	2	3676.76	6/28/2004 0:00	Shipped
15	10275	45	92.83	1	4177.35	7/23/2004 0:00	Shipped
16	10285	36	100	6	4099.68	8/27/2004 0:00	Shipped
17	10299	23	100	9	2597.39	9/30/2004 0:00	Shipped
18	10309	41	100	5	4394.38	10/15/2004 0:00	Shipped
19	10318	46	94.74	1	4358.04	11/2/2004 0:00	Shipped
20	10329	42	100	1	4396.14	11/15/2004 0:00	Shipped
21	10341	41	100	9	7737.93	11/24/2004 0:00	Shipped
22	10361	20	72.55	13	1451	12/17/2004 0:00	Shipped
23	10375	21	34.91	12	733.11	2/3/2005 0:00	Shipped
24	10388	42	76.36	4	3207.12	3/3/2005 0:00	Shipped
25	10403	24	100	7	2434.56	4/8/2005 0:00	Shipped
26	10417	66	100	2	7516.08	5/13/2005 0:00	Disputed
27	10103	26	100	11	5404.62	1/29/2003 0:00	Shipped
28							

25 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 12:55

DATA MODELING (STAR SCHEMA):

Star Schema

A Star Schema is a widely used data modeling technique in Business Intelligence. It consists of one central fact table connected to multiple dimension tables. This structure improves query performance and simplifies reporting.

Fact Table

- Fact_Sales (sales_data_sample)

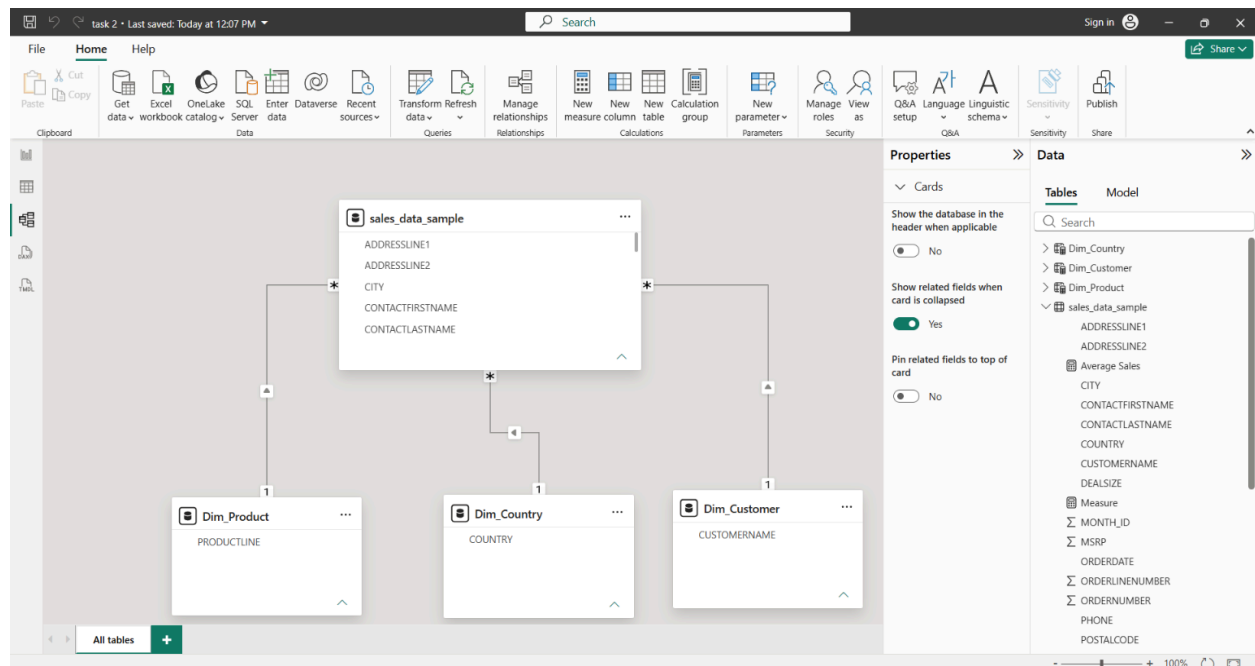
Dimension Tables

- Dim_Product
- Dim_Country
- Dim_Customer

Relationships

- Dim_Product → PRODUCTLINE → Fact_Sales
- Dim_Country → COUNTRY → Fact_Sales
- Dim_Customer → CUSTOMERNAME → Fact_Sales

The star schema enables efficient filtering and analysis while maintaining a simple and organized data model.



DAX MEASURES:

DAX Measures

Data Analysis Expressions (DAX) were used to create custom calculations for the dashboard.

Total Sales:

Total Sales = `SUM('sales_data_sample'[SALES])`

Calculates the overall sales generated in the dataset.

Total Orders:

Total Orders = `COUNT('sales_data_sample'[ORDERNUMBER])`

Calculates the total number of orders.

Total Customers:

Total Custome=`DISTINCTCOUNT('sales_data_sample'[CUSTOMERNAME])`

Calculates the number of unique customers.

Average Sales:

Average Sales = AVERAGE('sales_data_sample'[SALES])

Calculates the average sales amount.

DASHBOARD DEVELOPMENT:

The dashboard was developed using Microsoft Power BI to provide an interactive overview of sales performance.

The dashboard contains two report pages:

Dashboard Page

- Total Sales
- Total Orders
- Total Customers
- Average Sales
- Sales by Product Line
- Monthly Sales Trend

Sales Analysis Page

- Sales by Deal Size
- Sales by Order Status
- Sales by Country
- Interactive Slicers

The dashboard allows users to filter and analyze sales data efficiently.

DASHBOARD VISUALIZATIONS:

The following visualizations were created:

KPI Cards

KPI cards display important business metrics such as total sales, total customers, total orders, and average sales.

Bar Chart

The bar chart represents total sales across different product lines, helping identify the highest-performing products.

Line Chart

The line chart displays monthly sales trends, allowing users to observe sales growth and seasonal variations.

Donut Chart

The donut chart represents sales distribution according to deal size.

Pie Chart

The pie chart illustrates the distribution of sales based on order status.

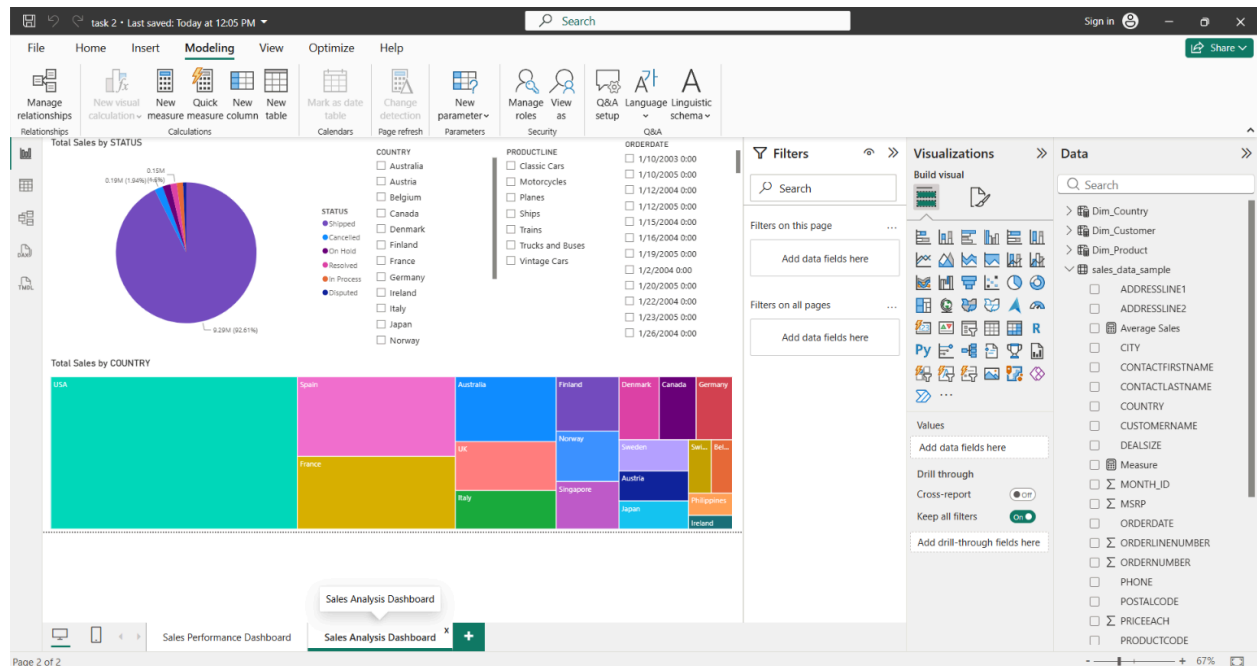
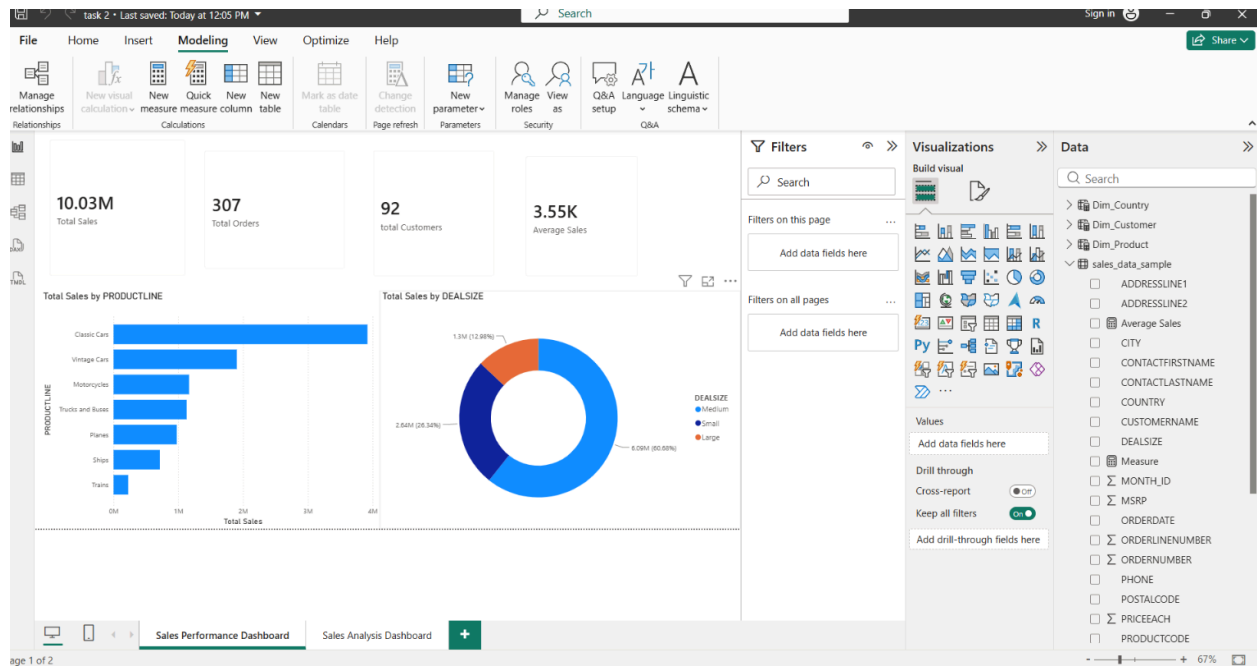
Treemap

The treemap visualizes sales distribution across different countries.

Slicers

Interactive slicers allow users to filter dashboard data by:

- Country
- Product Line
- Year



Key Insights:

The analysis of the dashboard provided the following business insights:

- Total sales indicate overall business performance.

- Certain product lines contribute significantly more revenue than others.
- Sales vary across different countries.
- Deal size influences total sales revenue.
- Monthly sales trends help identify seasonal demand.
- Interactive filtering allows users to explore business performance dynamically.

Challenges Faced:

During the project, several challenges were encountered:

- Learning Power BI for the first time.
- Understanding DAX calculations.
- Creating relationships between fact and dimension tables.
- Building an effective star schema.
- Designing a clean and interactive dashboard.
- Working with Power Query transformations.

These challenges were resolved through continuous practice and experimentation.

Conclusion:

This project successfully demonstrated the use of Microsoft Power BI for Business Intelligence and data visualization. The sales dataset was cleaned, transformed, modeled, and analyzed using Power Query, DAX measures, and interactive visualizations.

The developed dashboard enables users to understand sales performance, customer behavior, product trends, and geographical distribution of sales. The project also provided practical experience in data preprocessing, data modeling, and dashboard development.

Overall, the project enhanced practical knowledge of Business Intelligence concepts and Power BI development.